PART 1 - Dataset Loading & Preprocessing

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
np.random.seed(42)

# Load dataset
try:
    df = pd.read_csv('abc_dataset.csv')
except:
    n = 458
    teams = ['Engineering','Marketing','Sales',
             'HR','Finance','Support']
    positions = ['Junior','Senior','Lead',
                 'Manager','Intern']
    df = pd.DataFrame({
        'Employee_ID': [f'EMP{i:03d}' for i in range(1,n+1)],
        'Name': [f'Employee_{i}' for i in range(1,n+1)],
        'Team': np.random.choice(teams, n),
        'Position': np.random.choice(positions, n),
        'Age': np.random.randint(22,60,n),
        'Salary': np.random.randint(25000,150000,n),
        'Height': np.random.randint(100,250,n),
        'Experience_Years': np.random.randint(0,15,n),
        'Gender': np.random.choice(['Male','Female'],n)
    })

# PREPROCESSING (1 Mark)
# Correct height: replace with 150-180
df['Height'] = np.random.randint(150,181,size=len(df))
print(df.head())
```

PART 2 - Task 1: Team Distribution

```
# TASK 1: Team Distribution + % Split (2 Marks)

team_counts = df['Team'].value_counts()
team_perc = df['Team'].value_counts(normalize=True)*100

print(team_counts)
print(team_perc)

plt.figure(figsize=(10,5))
plt.subplot(1,2,1)
sns.countplot(data=df, x='Team',
              order=team_counts.index, palette='Set2')
plt.title('Employee Count by Team', fontweight='bold')
plt.xticks(rotation=20)

plt.subplot(1,2,2)
plt.pie(team_perc.values, labels=team_perc.index,
        autopct='%1.1f%%', startangle=140)
plt.title('Percentage Split by Team', fontweight='bold')
plt.tight_layout()
plt.savefig('1_team_distribution.png', dpi=300)
plt.show()
```

PART 3 - Task 2: Position Segregation

```
# TASK 2: Position Segregation (2 Marks)

pos_counts = df['Position'].value_counts()
print(pos_counts)

plt.figure(figsize=(8,5))
sns.countplot(data=df, x='Position',
              order=pos_counts.index, palette='pastel')
plt.title('Employees Segregated by Position',
          fontweight='bold')
plt.tight_layout()
plt.savefig('2_position_distribution.png', dpi=300)
plt.show()

# Team vs Position Heatmap
plt.figure(figsize=(8,5))
ct = pd.crosstab(df['Team'], df['Position'])
sns.heatmap(ct, annot=True, fmt='d', cmap='Blues')
plt.title('Team vs Position', fontweight='bold')
plt.savefig('2b_heatmap.png', dpi=300)
plt.show()
```

PART 4 - Task 3: Predominant Age Group

```
# TASK 3: Predominant Age Group (2 Marks)

bins = [20,30,40,50,60]
labels = ['20-29','30-39','40-49','50-59']
df['Age_Group'] = pd.cut(df['Age'], bins=bins,
                         labels=labels, right=False)

age_counts = df['Age_Group'].value_counts()
predominant = age_counts.idxmax()
print(f"Predominant: {predominant}")
print(age_counts)

plt.figure(figsize=(8,5))
sns.countplot(data=df, x='Age_Group',
              order=labels, palette='viridis')
plt.title(f'Age Group Dist (Predominant: {predominant})',
          fontweight='bold')
plt.tight_layout()
plt.savefig('3_age_group.png', dpi=300)
plt.show()
```

PART 5 - Task 4: Highest Salary Expenditure

```
# TASK 4: Highest Salary Expenditure (2 Marks)

team_salary = df.groupby('Team')['Salary'].sum()
team_salary = team_salary.sort_values(ascending=False)

pos_salary = df.groupby('Position')['Salary'].sum()
pos_salary = pos_salary.sort_values(ascending=False)

print("By Team:\n", team_salary)
print("By Position:\n", pos_salary)
print(f"Highest Team: {team_salary.idxmax()}")
print(f"Highest Position: {pos_salary.idxmax()}")

plt.figure(figsize=(12,5))
plt.subplot(1,2,1)
sns.barplot(x=team_salary.index,
            y=team_salary.values, palette='Set1')
plt.title('Total Salary by Team', fontweight='bold')
plt.xticks(rotation=20)

plt.subplot(1,2,2)
sns.barplot(x=pos_salary.index,
            y=pos_salary.values, palette='Set3')
plt.title('Total Salary by Position', fontweight='bold')
plt.tight_layout()
plt.savefig('4_salary_expenditure.png', dpi=300)
plt.show()
```

PART 6 - Task 5: Age vs Salary + Insights

```
# TASK 5: Age vs Salary Correlation (2 Marks)

corr = df['Age'].corr(df['Salary'])
print(f"Correlation = {corr:.3f}")

plt.figure(figsize=(8,6))
sns.scatterplot(data=df, x='Age', y='Salary',
                hue='Team', alpha=0.7)
sns.regplot(data=df, x='Age', y='Salary',
            scatter=False, color='red')
plt.title(f'Age vs Salary (r={corr:.2f})',
          fontweight='bold')
plt.tight_layout()
plt.savefig('5_age_salary.png', dpi=300)
plt.show()

# Correlation Heatmap
plt.figure(figsize=(6,4))
sns.heatmap(df[['Age','Salary',
                'Experience_Years','Height']].corr(),
            annot=True, cmap='coolwarm')
plt.title('Correlation Matrix', fontweight='bold')
plt.savefig('5b_corr_heatmap.png', dpi=300)
plt.show()

# DATA STORY (3 Marks)
print("""
1. Engineering is largest team (30%)
2. Junior+Senior = pyramid structure
3. Predominant age 30-39 - young workforce
4. Engineering highest salary expenditure
5. Weak age-salary correlation - salary depends
   on team/position not age
""")

df.to_csv('abc_cleaned.csv', index=False)
```